

**FIITJEE**  
**ALL INDIA TEST SERIES**  
**JEE (Advanced)-2024**  
**PART TEST – III**  
**PAPER –2**  
**TEST DATE: 24-12-2023**

**Time Allotted: 3 Hours**

**Maximum Marks: 180**

**General Instructions:**

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.  
**Section – A (01 – 04, 18 – 21, 35 – 38):** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.  
**Section – A (05 – 07, 22 – 24, 39 – 41):** This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).  
**Section – B (08 – 13, 25 – 30, 42 – 47):** This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.  
**Section – C (14 – 17, 31 – 34, 48 – 51):** This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

**MARKING SCHEME**

**Section – A (Single Correct):** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                                                     |
|----------------|---|----|---------------------------------------------------------------------|
| Full Marks     | : | +3 | If ONLY the correct option is chosen.                               |
| Zero Marks     | : | 0  | If none of the options is chosen (i.e. the question is unanswered); |
| Negative Marks | : | –1 | In all other cases.                                                 |

**Section – A (One or More than One Correct):** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                                                                                     |
|----------------|---|----|-----------------------------------------------------------------------------------------------------|
| Full Marks     | : | +4 | If only (all) the correct option(s) is (are) chosen;                                                |
| Partial Marks  | : | +3 | If all the four options are correct but ONLY three options are chosen;                              |
| Partial marks  | : | +2 | If three or more options are correct but ONLY two options are chosen and both of which are correct; |
| Partial Marks  | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option;        |
| Zero Marks     | : | 0  | If none of the options is chosen (i.e. the question is unanswered);                                 |
| Negative Marks | : | –2 | In all other cases.                                                                                 |

**Section – B:** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                         |
|----------------|---|----|-----------------------------------------|
| Full Marks     | : | +4 | If ONLY the correct integer is entered; |
| Zero Marks     | : | 0  | Question is unanswered;                 |
| Negative Marks | : | 0  | In all other cases.                     |

**Section – C:** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                         |
|----------------|---|----|-----------------------------------------|
| Full Marks     | : | +3 | If ONLY the correct integer is entered; |
| Zero Marks     | : | 0  | Question is unanswered;                 |
| Negative Marks | : | 0  | In all other cases.                     |

# Physics

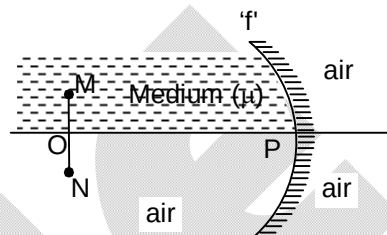
## PART – I

### SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

1. A medium of refractive index  $\mu = \frac{3}{2}$  is filled in front of concave mirror above the principal axis as shown in the figure. The radius of curvature of mirror is 30 cm. An object MN of length 2 mm is kept symmetrically on principal axis at a distance of 20 cm from the pole of mirror. Consider paraxial rays only. Let M' and N' are the images of point M and N formed by mirror then



- (A) M' will be in the medium and N' is the air, and distance between them is 12 mm.  
 (B) M' will be in the air and N' is the medium, and distance between them is 6 mm.  
 (C) M' will be in the medium and N' is the air, and distance between them is 6 mm.  
 (D) M' will be in the air and N' is the medium, and distance between them is 12 mm.
2. A plane electromagnetic wave is travelling in a space along positive y-axis. The total average energy density of wave in the space is  $\frac{8000}{\pi} \text{ J/m}^3$  and its frequency is given by  $1.8 \times 10^{11} \text{ Hz}$ .

The electric field and magnetic field in the space is best represented by

(A)  $\vec{E} = 1.2 \times 10^7 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](-\hat{i}) \text{ V/m}$

$\vec{B} = 0.04 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](+\hat{k}) \text{ Tesla}$

(B)  $\vec{E} = 1.2 \times 10^7 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](+\hat{i}) \text{ V/m}$

$\vec{B} = 0.04 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](+\hat{k}) \text{ Tesla}$

(C)  $\vec{E} = 2.4 \times 10^7 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](-\hat{i}) \text{ V/m}$

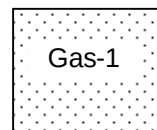
$\vec{B} = 0.08 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](+\hat{k}) \text{ Tesla}$

(D)  $\vec{E} = 2.4 \times 10^7 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](+\hat{i}) \text{ V/m}$

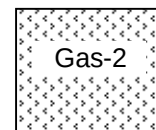
$\vec{B} = 0.08 \sin[(3.6\pi \times 10^{11})t - (1.2\pi \times 10^3)y](+\hat{k}) \text{ Tesla}$

where, y is in meter and 't' in second

3. A monoatomic gas-1 of mass  $m_1 = 5\text{gm}$  is kept in container-A and another monoatomic gas-2 of mass  $m_2 = 8\text{ gm}$  is kept in container-B. The speed of sound in gas-1 and gas-2 is  $v_1 = 400 \text{ m/s}$  and  $v_2 = 300 \text{ m/s}$  respectively at same temperature ' $T_0$ '. Now if both gases are mixed in another container without any leakage then speed of sound in mixture at same temperature  $T_0$  is



Container -A



Container -B

(A)  $100\sqrt{\frac{262}{27}} \text{ m/s}$

(B)  $100\sqrt{\frac{44}{13}} \text{ m/s}$

(C)  $100\sqrt{\frac{152}{13}} \text{ m/s}$

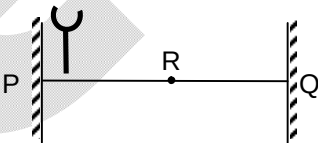
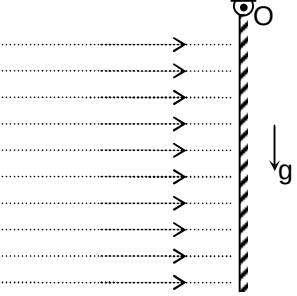
(D)  $100\sqrt{\frac{52}{27}} \text{ m/s}$

4. A nucleus of Uranium  $^{238}\text{U}$  undergoes a reaction as given below. It converted into the thorium  $^{234}\text{Th}$  and  $\alpha$ -particle  $^4\text{He}$
- $$^{238}\text{U} \rightarrow ^{234}\text{Th} + ^4\text{He}$$
- The atomic masses of nucleus are  
 $m_{\text{U}} = 238.05079 \text{ amu}$   
 $m_{\text{Th}} = 234.04363 \text{ amu}$   
 $m_{\text{He}} = 4.00260 \text{ amu}$   
 (given  $1 \text{ amu} = 1.67 \times 10^{-27} \text{ kg}$ )  
 The maximum kinetic energy of  $\alpha$ -particle is
- (A)  $3.2 \times 10^{-13} \text{ J}$  (B)  $9.2 \times 10^{-13} \text{ J}$   
 (C)  $1.5 \times 10^{-13} \text{ J}$  (D)  $6.8 \times 10^{-13} \text{ J}$

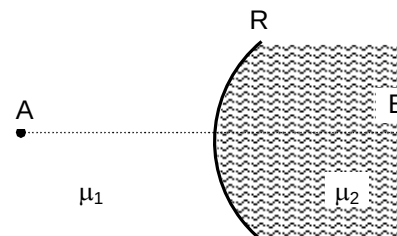
### SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A string of uniform mass density is fixed between two points P and Q. A tuning fork is used to set up a standing wave in the string. It is observed that mid point 'R' of string AB is node. Tension in the string is uniform. Regarding the standing wave in string choose correct option(s).
- 
- (A) There may be a phase difference of  $\frac{\pi}{3}$  between two point on the string.  
 (B) Total mechanical energy of a considered small element remain constant during the motion of particle.  
 (C) There can be two points between P and R, which will have equal speed at all times.  
 (D) When particles reaches their extreme position then particle near the node has minimum potential energy.
6. A plane mirror hangs from point 'O' in vertical plane. Mirror is free to rotate about point O and frictional force is negligible between hinge and mirror. Now a monochromatic light is incident on plane mirror normally. Mirror does not behave like a perfectly absorbing surface, incident light gets reflected also. Choose the correct option(s).
- 
- (A) Wave length of reflected light is more than that of incident light.  
 (B) If reflectivity of mirror is one then radiation pressure will be  $\frac{2I}{C}$  at  $t = 0$ . Where I is intensity of incident light and 'C' is speed of light.  
 (C) Mirror will start to oscillate  
 (D) The angular displacement of mirror does not depends on wavelength of incident light

7. There are two different medium of refractive indices  $\mu_1$  and  $\mu_2$  which are separated by a curved spherical surface of radius of curvature 'R' as shown in the figure. The incident ray of an object lies in medium  $\mu_1$ . It is given that  $\mu_1 > 0$  and  $\mu_2 > 0$  and AB is the principal axis of curved surface. Regarding the position of object and image, choose the correct option(s)



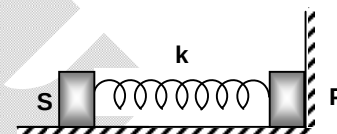
- (A) If  $\mu_2 > \mu_1$ , then there must be virtual image of virtual object  
 (B) If  $\mu_2 > \mu_1$ , then there must be real image of virtual object  
 (C) If  $\mu_1 > \mu_2$ , then there must be virtual image of real object  
 (D) If  $\mu_1 > \mu_2$ , then there must be real image of real object

### SECTION – B

#### (Numerical Answer Type)

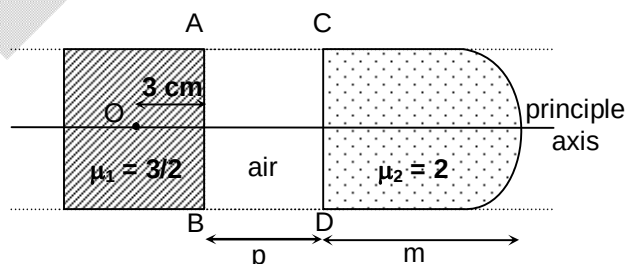
This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

8. Two small similar charged blocks are connected from ends of spring of force constant  $k = 420 \text{ N/m}$ . The charges on the blocks is  $q_0 = 100 \mu\text{C}$ . The block 'P' is fixed and S can move on frictionless horizontal surface. The mass of each block is  $m = 60 \text{ gm}$ . In equilibrium position the distance between P and S is  $L = 1 \text{ m}$ . Now block 'S' is slightly displaced along the length of spring from its equilibrium position. Find angular frequency of oscillation of S.

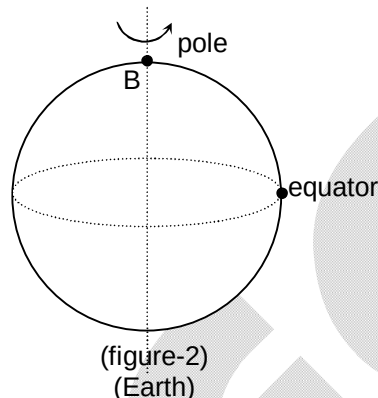
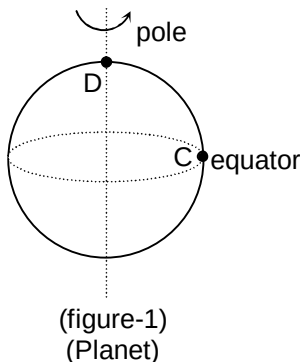


9. Consider a metallic bottle of volume 4 litre. Air of density  $1.5 \text{ kg/m}^3$  is pumped upto a pressure of 3 atm inside the bottle. There is a large plastic bag of negligible thickness with no air inside it and capacity more than 16 litre. Now, bag is connected to the opening of the closed bottle and is placed on one pan of balance and weights are put on the other pan to establish balance. When the opening of the bottle is opened air flows from bottle into the bag so that balance is disturbed. Assuming temperature of gas remain constant. Find additional weight (in gm) which is required to reestablish the balance.

10. A luminous point object 'O' is kept inside a glass slab of refractive index  $\mu_1 = \frac{3}{2}$  at a distance 3 cm from its one surface AB. A thick plano-convex lens of thickness  $m \text{ cm}$  is kept at a distance of  $p \text{ cm}$  from surface AB. The refractive index of lens is 2 and its radius of curved surface is 15 cm. The width of air gap between glass slab and lens is  $p$  which can be adjusted by moving glass slab and lens. If light rays after passing through lens are always parallel to principal axis of lens means final image is formed at infinity, then find maximum width  $m$  in cm of plano-convex lens.



11. A spherical planet of uniform mass density has radius one third of the radius of the Earth. A man experiences his weight at 'D' which is one fourth of the weight that he observed at B. He also found himself to be weightless at point C. The mass density of the earth is ' $\rho$ '. The duration of a day on the planet is given by  $\sqrt{\frac{\lambda_1}{\lambda_2 G \rho}}$ . Find the value of  $\lambda_1 + \lambda_2$ . (Take,  $\pi = \frac{22}{7}$ )



12. A monochromatic point source 'S' of wavelength  $\lambda = 4200 \text{ \AA}$  (in air) is used in YDSE as shown in the figure-1. The distance between slits  $S_1$  and  $S_2$  is  $d = 1 \text{ mm}$  and screen is at a distance  $D = 2 \text{ m}$  from the slits. Sheet-2 is kept back to slit  $S_2$  and having refractive index  $\mu_2 = \frac{4}{3}$  and thickness  $t_0 = 1 \text{ }\mu\text{m}$ . Sheet -1 is kept in front of slit  $S_1$  and its refractive index ( $\mu_1$ ) varies as shown in the figure-2. Left of slits  $S_1 S_2$ , a medium of refractive index  $\mu_3 = \frac{3}{2}$  is present and to the right of slits  $S_1 S_2$ , a medium of refractive index  $\mu_4 = 2$  is present. For central maxima to be at 'O', the required thickness 't' of sheet-1 is  $\frac{5}{k} \text{ }\mu\text{m}$ . Find the value of k.

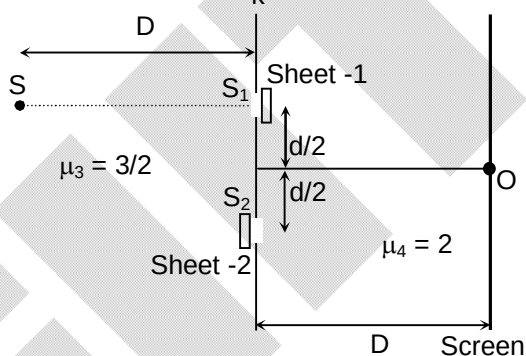


Figure-1

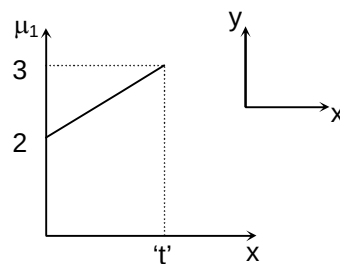
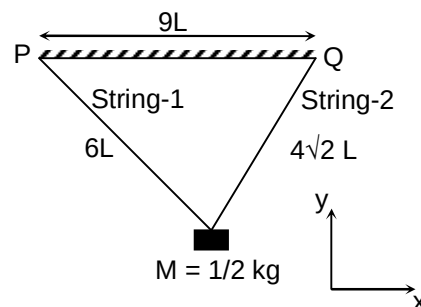


Figure-2

13. A block of mass  $M = \frac{1}{2} \text{ kg}$  is connected with two massless string of length  $6L$  and  $4\sqrt{2}L$ . The other ends of strings are fixed at P and Q in a vertical plane as shown in the figure. Now block is slightly displaced along z-axis and then released. The time period of oscillation of block is given by  $T = 2\pi\sqrt{\frac{\lambda_1}{\lambda_2}}$ . It is given that  $L = 2 \text{ m}$  and  $g = 10 \text{ m/s}^2$ . Find the value of  $(\lambda_1 + \lambda_2)$ .



### SECTION – C

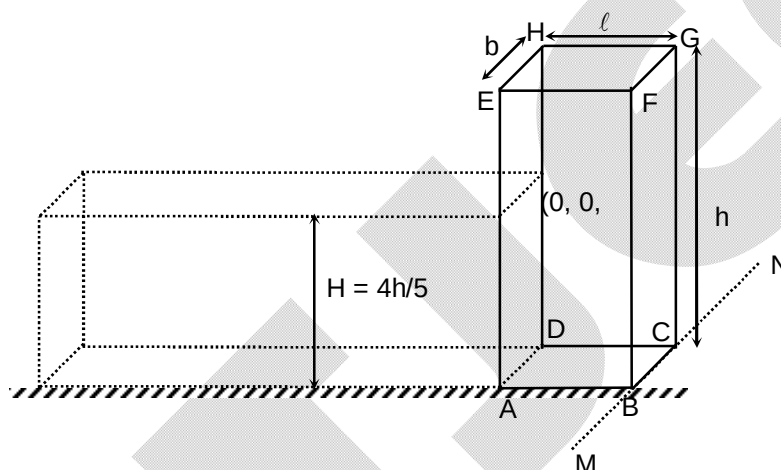
(Numerical Answer Type)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

#### Paragraph for Question Nos. 14 and 15

ABCDEFGH is cuboidal concrete block as shown in the figure. It is used as a retaining wall in a reservoir of water. The relative density of block is  $\frac{16}{5}$ . The block can not slide but can rotate about a line MN.

There is no seepage of water on the face ABFE and CGHD but there is a seepage and a thin film of water is formed under the face ABCD. Assume that a seal on line MN is present to prevent the water from flowing out underneath the block. The height of water is  $H = \frac{4h}{5}$ . (Given  $\sqrt{15} = 3.88$ )



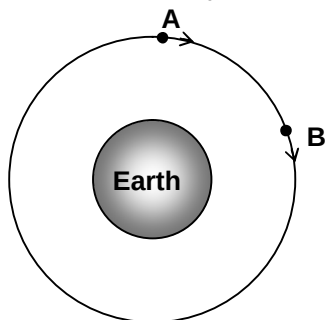
- 14 The minimum value of  $\left(\frac{\ell}{h}\right)$  for which the cuboidal concrete block will not topple about line MN is .....

15. The ratio of torques on the face ABCD and the face ADHE about line MN of the cuboidal concrete block due to water is .....

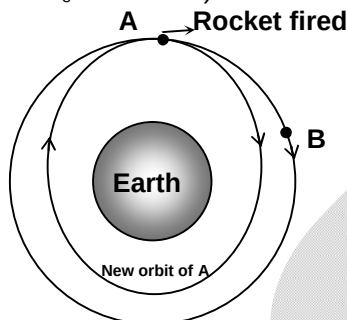
(Take the value of  $\left(\frac{\ell}{h}\right)$  as obtained in the first part)

**Paragraph for Question Nos. 16 and 17**

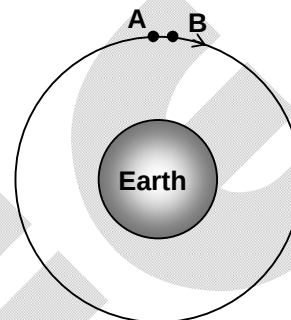
Two artificial satellites A and B of masses  $m_A = 300 \text{ kg}$  and  $m_B = 3000 \text{ kg}$  are orbiting in circular orbit around the Earth as shown in the figure. Earth is at rest. The initial orbit radius of A and B is  $r = 9000 \text{ km}$ . Satellite B is ahead of A by a time  $\Delta T$ . A rocket is fired from satellite 'A' in such a way that its speed decreases instantly by  $0.6\%$  as a result it moves in elliptical orbit as shown in figure-2. The new path traced by A is in a way such that it coincides with B at the initial position of A in one rotation as shown in figure-3. (Assuming Newton's law of gravitation and Kepler's law is valid for satellite A and B) (take  $\pi = 3.14$ ,  $G = 6.67 \times 10^{-11}$ ,  $M_e = 6 \times 10^{24} \text{ kg}$  and  $R_e = 6400 \text{ km}$ )



**Figure-1**



**Figure-2**



**Figure-3**

16. The new semi major axis of satellite A .....in km.
17. The value of  $\Delta T$  .....in sec

# Chemistry

## PART – II

### SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

18. Which of the following is correct regarding the drug 'Valium'?
- It is an antipyretic medicine.
  - It contains two 'Cl' atoms per molecule.
  - Its functional group can be tested by  $\text{AgNO}_3$  (aq) to obtain a white precipitate.
  - It contains two 'N' atoms per molecule.
19. What is the value of  $E_{\text{cell}}$  of the following, at  $25^\circ\text{C}$ ?
- $\text{Pt}; \text{Cl}_2 (1 \text{ atm}) | \text{Cl}^- (0.1 \text{ M}) || \text{Cr}_2\text{O}_7^{2-} (0.01 \text{ M}), \text{Cr}^{3+} (0.1 \text{ M}) (\text{in } 0.05 \text{ M H}_2\text{SO}_4)$
- Given  $E_{\text{Cr}_2\text{O}_7^{2-}|\text{Cr}^{3+}}^\circ = 1.23 \text{ V}$ ;  $E_{\text{Cl}^-|\text{Cl}_2}^\circ = -1.36 \text{ V}$
- $-0.371 \text{ V}$
  - $-0.237 \text{ V}$
  - $-0.327 \text{ V}$
  - $+0.237 \text{ V}$
20. Two inorganic salt solutions of Na are contained in two beakers X and Y. Both are clear and colourless solutions. A few amounts of solution are taken in two different test tubes separately, and treated with  $\text{K}_2\text{Cr}_2\text{O}_7$  and conc.  $\text{H}_2\text{SO}_4$  and both produce the reddish vapours. The gases/vapours evolved are passed into NaOH solutions separately and it was observed that the vapours evolved from sample taken from 'X', imparted yellow colour to NaOH solution, while other solution remains colourless, on passing the vapours evolved from the sample taken from 'Y'. Which of the following is correct regarding the above experiment?
- Beaker 'X' contains  $\text{Na}_2\text{CrO}_4$
  - Beaker 'X', contains NaBr
  - Beaker 'Y' contains NaCl
  - Reddish vapours evolved from the sample taken from 'Y' are  $\text{Br}_2$  vapours
21. A solid is formed and constituted by three elements 'X', 'Y' and 'Z'. X form simple cubic unit cells while Y atoms occupy all the face centres while Z occupy body centres of each cubic unit cell. If any one corner particle is missing from each of the unit cell corresponding, then what is the formula of the compound?
- $\text{X}_3\text{Y}_{12}\text{Z}_4$
  - $\text{X}_7\text{Y}_{16}\text{Z}_8$
  - $\text{X}_3\text{Y}_2\text{Z}$
  - $\text{X}_7\text{Y}_{24}\text{Z}_8$

### SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

22. Which of the following is/are correct for an ideal gas undergoing isothermal process? [All notations have their usual meaning]
- $\left(\frac{\partial G}{\partial P}\right)_T = +V$
  - $\left(\frac{\partial A}{\partial V}\right)_T = -P$
  - $\Delta H = 0$
  - $w = q$



23. Which of the following form(s) cyclic trimer or tetramer structures?  
 (A)  $\text{SO}_3$  (B)  $\text{SeO}_3$   
 (C)  $\text{ClO}_2$  (D)  $\text{HPO}_3$
24. Which of the following give(s) white precipitate on addition to  $\text{Na}_2\text{CO}_3$  solution?  
 (A)  $\text{BaCl}_2$  (B)  $\text{CaCl}_2$   
 (C)  $\text{AgNO}_3$  (D)  $\text{Cu}(\text{NO}_3)_2$

### SECTION – B

#### (Numerical Answer Type)

*This section contains SIX (06) Numerical based questions. The answer to each question is a NON-NEGATIVE INTEGER VALUE.*

25. A salt AB crystallizes into a cubic lattice. Radii of  $\text{A}^+$  and  $\text{B}^-$  are 181 pm and 226 pm, respectively. What is the distance between second nearest neighbours? [Assume most suitable crystal lattice formation]
26. What is the numerical value of heat of formation of  $\text{CS}_2(\ell)$  (in kJ/mole)?  
 Given: Heat of formation of  $\text{CO}_2(\text{g}) = -393.5$  kJ/mole  
 Heat of formation of  $\text{SO}_2(\text{g}) = -293.72$  kJ/mole  
 and Heat of combustion of  $\text{CS}_2(\ell) = -1107.94$  kJ/mole
27. 11.4 gm mixture of  $\text{Na}_2\text{CO}_3$  and  $\text{NaHCO}_3$  is dissolved into 2 litre water. 50 ml of the solution require 21 ml, 0.1 M HCl in presence of phenolphthalein for first end point. What would be amount of  $\text{CO}_2$  released (in gm), if excess of HCl is added to 2 litre original solution?
28. Upon heating 204.17 gm  $\text{KClO}_3$  in presence of catalytic amount of  $\text{MnO}_2$ , a gas 'P' is formed. The same amount of 'P' reacts with sufficient amount of white phosphorous to give 'Q'. What is the weight of 'Q' formed (in gm)? [At. Wt. of P = 31, O = 16, Cl = 35.5, K = 39]
29. Consider the following reactions:  
 (i)  $\text{NH}_4\text{Cl} \xrightarrow[\Delta]{\text{NaOH}} \text{X}(\text{g});$  (X contains nitrogen)  
 (ii)  $\text{PH}_4\text{I} \xrightarrow[\Delta]{\text{NaOH}} \text{Y}(\text{g});$  (Y contains phosphorus)  
 (iii)  $\text{P}_4 \xrightarrow[\Delta]{\text{NaOH}} \text{Z}(\text{s});$  (Z contains phosphorus in +1 oxidation state)  
 What is the sum of molar masses of X, Y and Z?
30. A solution of a purple coloured complex with the molecular formula  $\text{CoCl}_3 \cdot \text{XNH}_3$  is treated with sufficient amount of  $\text{AgNO}_3$  solution. How much amount (in gm) of white precipitate will be obtained if 1 mole of the said complex was dissolved in the solution?

### SECTION – C (Numerical Answer Type)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

#### Paragraph for Question Nos. 31 and 32

A metal 'M' is boiled in dilute  $\text{HNO}_3$  to give salt 'N' and a natural oxide of nitrogen (O). An aqueous solution of 'N' with brine solution gives a white precipitate 'P'. When 'N' was heated with  $\text{Na}_2\text{S}_2\text{O}_3$  solution a white precipitate 'Q' was obtained which on standing turns to a black compound 'R'.

31. What is the molar mass of the gas 'O'?
32. What amount (in grams) of 'Q' is formed if 54 grams of the metal 'M' is taken in the process and the yield is 100% and also total amount of solution of 'N' is treated with excess  $\text{Na}_2\text{S}_2\text{O}_3$ ?

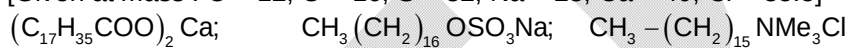
#### Paragraph for Question Nos. 33 and 34

There are three types of detergents

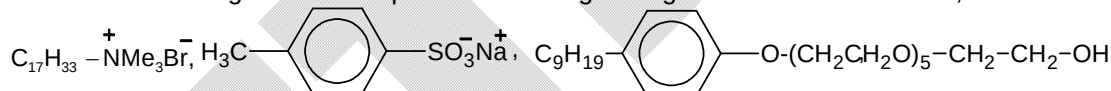
1. Cationic – which are acetates or chlorides of  $4^\circ$  amines.
2. Anionic – which contain long chain alkyl sulphates.
3. Non-ionic – which are formed by polyethylene glycols and stearic acids.

33. One of the following not used in synthetic detergent. What is the molar mass of that one?

[Given at mass : C = 12, O = 16, S = 32, Na = 23, Ca = 40, Cl = 35.5]



34. One of the following is used in liquid dish washing detergent with molar mass 'M';



What is the value of  $\frac{M}{4}$ ?

[Given at mass : C = 12, N = 14, O = 16, Br = 80, H = 1]

# Mathematics

## PART – III

### SECTION – A

(One Options Correct Type)

This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

35. For all complex number  $z$  lying on a unit circle in complex plane centred at origin, the range of  $f(z) = |z^2 + 1 - z| + |z + 1|$  is  
 (A)  $\left[\frac{3}{4}, \sqrt{3}\right]$  (B)  $\left[0, \frac{13}{4}\right]$   
 (C)  $\left[\sqrt{3}, \frac{13}{4}\right]$  (D)  $[0, \sqrt{3}]$
36. If  $p$  and  $q$  be the first of  $n$  A.M.s and  $n$  H.M.s between any two real numbers, then  
 (A)  $\frac{q}{p}$  can lie between 1 and  $\left(\frac{n+1}{n-1}\right)^2$  (B)  $\frac{q}{p}$  can't lie between  $\left(\frac{n-1}{n+1}\right)^2$  and 1  
 (C)  $\frac{q}{p}$  can't lie between 1 and  $\left(\frac{n+1}{n-1}\right)^2$  (D)  $\frac{q}{p}$  can't lie between 1 and  $\frac{n^2-1}{n^2+1}$
37. Let  $f(x) = \left(\frac{p^2-9}{p^2+3}\right)x^3 - 3x + \tan 3$ , then roots of the equation  $\frac{1}{x-f(1)} - \frac{2}{x-f(2)} - \frac{3}{x-f(3)} = 0$  necessarily has  
 (A) at least one root between  $f(1)$  and  $f(2)$  when  $|p| < 3$   
 (B) at least one root between  $f(2)$  and  $f(3)$  when  $|p| < 3$   
 (C) at least one root between  $f(2)$  and  $f(3)$  when  $|p| > 3$   
 (D) at least one root between  $f(2)$  and  $f(3)$  for all real values of  $p$
38. Two distinct dices are loaded in such a way that every sum of their outcomes (2, 3, ..., 12) are equally likely. If the dices are loaded such that for first dice  $P(i \text{ appears}) \propto i^{\lambda_1}$  and for second dice  $P(i \text{ appears}) \propto i^{\lambda_2}$  (where  $i = 1, 2, 3, 4, 5, 6$ ), then  
 (A)  $\lambda_1 = \lambda_2$  (B)  $\lambda_1 \neq \lambda_2$   
 (C)  $\lambda_1 = \lambda_2 \in \mathbb{Q}$  (D) dices can't be loaded in such a manner

### SECTION – A

(One or More than one correct type)

This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Given  $A_\lambda = \begin{bmatrix} C_\lambda & 0 \\ 0 & C_{\lambda-1} \end{bmatrix}$  for  $\lambda \geq 1$ , where  $C_\lambda = {}^nC_\lambda$ ,  $0 \leq \lambda \leq n$  and  
 $A_1 A_2 - A_2 A_3 + A_3 A_4 - \dots + (-1)^{n-1} A_{n-1} A_n = \begin{bmatrix} p & 0 \\ 0 & q \end{bmatrix}$ , then  
 (A)  $p = n$ , if  $n$  is even (B)  $p = n - (-1)^{\frac{n-1}{2}} \cdot \frac{{}^nC_{n-1}}{2}$ , if  $n$  is odd  
 (C)  $q = -n$ , if  $n$  is even (D)  $q = \frac{{}^nC_{n-1}}{2} (-1)^{\frac{n-1}{2}} - (-1)^{n-1} \cdot n$ , if  $n$  is odd

40. If the straight lines  $\vec{r} = \lambda_1(\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ ,  $\vec{r} = \lambda_2(\ell\hat{i} + m\hat{j} + n\hat{k})$ ,  $\vec{r} = \lambda_3(3\alpha\hat{i} + 7\beta\hat{j} + 5\gamma\hat{k})$  are coplanar and  $k_1\frac{\ell}{\alpha} + \frac{m}{\beta} = k_2\frac{n}{\gamma}$ , then which of the following is/are CORRECT?
- (A)  $k_1^2 + k_2^3$  is a perfect cube  
(B)  $k_1^2 + k_2^3$  is a perfect square  
(C)  $k_1^2 + k_2^2$  is prime  
(D)  $k_1^2 + k_2^2$  is composite
41. If volume of a tetrahedron OABC (where O is origin) is 5 units and position vector of circumcentre of tetrahedron is given by  $\vec{r} = \frac{(\text{OA})^2\vec{r}_1 + (\text{OB})^2\vec{r}_2 + (\text{OC})^2\vec{r}_3}{120}$  and  $[\vec{r}_1 \vec{r}_2 \vec{r}_3] = N$ , then
- (A) sum of digits of N is a perfect square  
(B) sum of digits of N is a perfect cube  
(C) sum of digits of N is even  
(D) sum of digits of N is odd

### SECTION – B

#### (Numerical Answer Type)

This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.

42. The number of ways to perfectly cover a  $2 \times 8$  board with dominoes (domino means a tile of size  $2 \times 1$ ) is
43. The greatest term independent of x in the expansion of  $\left[x \cos \theta + \frac{\sin \theta}{x}\right]^{30}$  as  $\theta$  is real is  ${}^{30}C_\alpha(\beta)^{-\gamma}$ , where  $\alpha, \beta, \gamma \in \mathbb{N}$ , then  $\alpha + \beta + \gamma$  is
44. Let  $S = \sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} + \sqrt{1 + \frac{1}{3^2} + \frac{1}{4^2}} + \dots + \sqrt{1 + \frac{1}{2024^2} + \frac{1}{2025^2}}$ , then  $\frac{(2025)S - 2024}{2024}$  is
45. Consider the  $12 \times 12$  square matrix filled with natural numbers from 1 to 144 as given:
- |          |          |          |     |          |
|----------|----------|----------|-----|----------|
| 1        | 13       | 25       | ... | 133      |
| 2        | 14       | 26       | ... | 134      |
| 3        | 15       | 27       | ... | 135      |
| 4        | 26       | 28       | ... | 136      |
| $\vdots$ | $\vdots$ | $\vdots$ | ... | $\vdots$ |
| 12       | 24       | 36       | ... | 144      |
- A number is selected at random from the board and the corresponding row and column are deleted. The process is continued upto 12 times so that no row and no column is left. Then the sum of the numbers so selected is
46. If  $z_1, z_2, z_3$  are three complex numbers such that  $|z_1| = |z_2| = |z_3| = 1$  and  $\frac{z_1^2}{z_2 z_3} + \frac{z_2^2}{z_3 z_1} + \frac{z_3^2}{z_1 z_2} = -1$ , then the sum of possible values of  $|z_1 + z_2 + z_3|$  is

47. There are 6 Indian and 8 Australian players available. 5 players are selected at random and made to wear Indian team jersey. The remaining 9 are made to wear Australian team jersey. If the probability that the number of Indian players wearing Australian jersey plus the number of Australian players wearing Indian jersey is not a prime number is  $\frac{m}{n}$  ( $m$  and  $n$  are coprime natural numbers), then  $m + n$  is

**SECTION – C**  
**(Numerical Answer Type)**

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX)**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

**Paragraph for Question Nos. 48 and 49**

A random variable  $X$  takes value  $x$  with probability proportional to  $y = f(x)$  where  $x$  takes all non-negative integral values and  $f(x)$  satisfies:  $dy + x5^{-x} \ln 5 \, dx = 5^{-x} (1 - \ln 5) \, dx$  and  $f(0) = 1$ , then

48.  $10P(X \leq 1)$  equals
49.  $100P(X \geq 3)$  equals

**Paragraph for Question Nos. 50 and 51**

For any positive integer  $k$ , let  $g(k) = \frac{4k + \sqrt{4k^2 - 1}}{\sqrt{2k+1} + \sqrt{2k-1}}$ , then

50.  $\sum_{k=1}^{60} g(k)$  is equal to
51. If  $\lim_{k \rightarrow \infty} (k^a \cdot g(k))$  is finite and equal to  $L$ , then  $a + L$  is equal to